

Zinc

Fact Sheet for Health Professionals

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This is a fact sheet intended for health professionals. For a general overview, see our [consumer fact sheet](#).

Introduction

Zinc, an essential mineral, is naturally present in some foods, added to others, and available as a dietary supplement. Zinc is also found in some cold lozenges, over-the-counter drugs sold as cold remedies, and some denture adhesive creams.

Zinc is involved in many aspects of cellular metabolism. It is required for the catalytic activity of hundreds of enzymes, and it plays a role in enhancing immune function, protein and DNA synthesis, wound healing, and cell signaling and division [1-4]. Zinc also supports healthy growth and development during pregnancy, infancy, childhood, and adolescence and is involved in the sense of taste [2,3,5].

The total amount of zinc in the body is approximately 1.5 grams (g) in women and 2.5 g in men [2]. Most of this zinc is stored in skeletal muscle and bone [1-3].

Zinc homeostasis is maintained through absorption of zinc from the diet, excretion into the gastrointestinal tract, and reabsorption in the gastrointestinal lumen [2,3]. In general, as zinc intakes rise, the amount of zinc absorbed also increases, but its fractional absorption drops [2,3].

Assessing zinc status

Serum or plasma zinc concentrations are often used to assess zinc status, and concentrations in healthy people typically range from 80 to 120 mcg/dL. However, these measures have important limitations. For example, zinc concentrations in serum can be affected by age, sex, and time of day, and they do not always correlate with dietary or supplemental zinc intakes.

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Recommended Intakes

The Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine has established Recommended Dietary Allowances and Adequate Intakes for zinc. These values range from 8 to 12 mg for adults and from 2 to 13 mg for infants, children, and adolescents, depending on age, sex, and life stage.

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Sources of Zinc

Food

Meat, fish and other seafood, eggs, and dairy products are good sources of zinc. Oysters contain particularly high amounts of zinc. Beans, nuts, and whole grains also contain zinc, but the bioavailability of zinc from these foods is lower than that of animal foods.

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Dietary supplements

Forms of zinc that are commonly found in dietary supplements include zinc sulfate, zinc acetate, and zinc gluconate. Taking supplements that contain a substantial amount of iron (≥ 25 mg) at the same time as zinc supplements can reduce zinc absorption and plasma concentrations of zinc.

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Other sources

Some homeopathic products and nasal sprays contain zinc. These products are often sold over the counter for managing the symptoms of the common cold and flu and providing immune support. Zinc is also present in some denture creams. Although the zinc intakes from these products are not a concern when they are used as directed, excessive use could lead to zinc toxicity.

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Zinc Intakes and Status

While people in certain age groups, such as older adults, are at higher risk of having marginal zinc intakes, most people in the United States consume an adequate amount of zinc. Zinc deficiency is much more common in low-income and middle-income countries than in the United States.

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Zinc Deficiency

Zinc deficiency affects many different organs and tissues in the body, and the signs and symptoms can vary by age. For example, zinc deficiency can delay growth and cause diarrhea and alopecia in children, and it can alter cognitive and psychological function in older adults. In pregnant women, zinc deficiency can increase the risk of maternal morbidity and adverse birth outcomes.

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Groups at Risk of Zinc Inadequacy

Certain groups of people are more likely than others to have zinc inadequacy. These include people with gastrointestinal disorders and those who have had bariatric surgery, people who follow vegetarian or vegan diets, and women who are pregnant or lactating. In addition, infants who are exclusively breastfed after 6 months of age and children with sickle cell disease have a higher risk of zinc inadequacy, as do people with alcohol use disorder.

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Zinc and Health

This section focuses on the following diseases and disorders in which zinc might play a role: the common cold, pneumonia in children, HIV in children and adults, acute childhood diarrhea (which often results from infections), age-related macular degeneration (AMD), and type 2 diabetes. For information on zinc and COVID-19, see [Dietary Supplements in the Time of COVID-19](#).

The common cold

The clinical trials that have evaluated the use of zinc supplements in people with the common cold have reported mixed results. In general, studies have found that taking supplemental zinc in lozenge or syrup form early in the course of the cold can reduce the duration of the signs and symptoms of the cold. However, the use of these products does not appear to affect the severity of those signs and symptoms.

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Pneumonia in children

Because poor zinc status is associated with a higher risk of pneumonia in children, researchers have evaluated whether zinc supplementation can help prevent or treat this disease. Research suggests that zinc supplementation can reduce the incidence and prevalence of pneumonia in infants and children in low-income countries where zinc deficiency is common, but most research has found that the adjunctive use of zinc supplements to treat pneumonia in children does not significantly reduce the risk of death or the time to recovery.

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HIV in children and adults

People with HIV often have low zinc status. However, research on the use of zinc supplementation in people with HIV has produced mixed results. While some clinical trials have reported that zinc supplements have some benefits in people with HIV, such as reduced rates of immunological failure events and diarrhea, other research has found little to no effect on CD4+ T-cell counts, viral load, mortality, or the frequency of diarrhea.

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Childhood acute diarrhea

The results of several clinical trials indicate that zinc supplementation can reduce the duration of acute diarrhea in children who live in low-income countries. The World Health Organization and UNICEF recommend short-term zinc supplementation of 10 or 20 mg/day (depending on age) for 10 to 14 days to treat acute childhood diarrhea. However, because most trials that evaluated the use of zinc supplementation for diarrhea have been conducted in low-income countries where zinc deficiency is common, the effects may be marginal in well-nourished children.

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Age-related macular degeneration

Observational evidence suggests that older adults who have higher zinc intakes have a lower risk of age-related macular degeneration. The results from large clinical trials indicate that taking dietary supplements that contain zinc, copper, and certain antioxidants (known as AREDS formulations) slows the progression of age-related macular degeneration among people who are at high risk of developing the advanced form of this condition.

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Type 2 diabetes

Zinc depletion may play a role in the progression of type 2 diabetes, as some observational studies have suggested a link between low zinc intakes and a higher risk of this condition. In addition, some clinical trials have reported that zinc supplements have beneficial effects on certain outcomes, including total cholesterol, triglyceride levels, fasting glucose, and insulin resistance. However, other trials have had conflicting results regarding zinc's effects. Overall, there is insufficient evidence to support the use of zinc supplementation to reduce the risk of type 2 diabetes or mitigate its symptoms.

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Health Risks from Excessive Zinc

Getting excessive amounts of zinc can cause nausea, dizziness, headaches, gastric distress, vomiting, and loss of appetite. Doses of 50 mg of zinc or more over a period of weeks can inhibit copper absorption, reduce immune function, and lower high-density lipoprotein cholesterol levels; however, it is unlikely that a person would obtain this much zinc from food alone. The Tolerable Upper Intake Level for zinc is 40 mg for adults, and it ranges from 4 to 34 mg for infants, children, and adolescents, depending on age.

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Interactions with Medications

Zinc may interact with certain medications, such as quinolone antibiotics, tetracycline antibiotics, and penicillamine. In addition, some diuretics can decrease serum zinc concentrations by increasing zinc excretion in urine.

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Zinc and Healthful Diets

In general, a person's nutritional needs should be met primarily through the diet, including fortified foods. Dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone, especially during certain life stages. The *Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

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